

CMS Open Data H to tau tau Search: Phase 4c Full Observed Results

Analysis my_analysis

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Change Log

Phase 4c v1

- Applied the frozen Phase 4a/4b score-template method to full Run2012B/C TauPlusX data.
- Derived the full high-mT W+jets scale 0.8528 ± 0.0370 .
- Preserved the Phase 4b score-modeling warning in the observed result.
- Recorded that the 4b human gate was auto-passed by explicit user instruction.

1 Phase 4c Observed Inference: Full Data

1.1 Summary

Phase 4c applies the frozen Phase 4a/4b score-template model to all available Run2012B/C TauPlusX data in `phase3_selection/outputs/sensitivity_selected_events.npz`. No full-data retuning of the selection, classifier, categories, score bins, or statistical model was performed. The Phase 4b human gate is recorded as auto-passed by explicit user instruction.

The fitted observable remains `mva_score_hist_gradient_boosting` in the `vbf`, `boosted`, and `zero_jet` channels with bin edges `[0.0, 0.2, 0.35, 0.5, 1.0]`. The Phase 4b score-template warning is not removed: the 10% combined data/MC ratio was 1.232, `chi2/ndf` was 1.732, and the status was `flagged` with flags `{'category_chi2_failure': False, 'combined_chi2_failure': False, 'combined_ratio_failure': True}`.

1.2 W High-mT Control Scale

The full-data W+jets scale is derived only from the high-mT control region using $(N_{\text{data_CR}} - \text{nonW_MC_CR}) / W_MC_CR$. The full control region contains 4023 data events, `W_MC` = 2414.781, and `nonW_MC` = 1963.570. The applied full-data scale is 0.8528 ± 0.0370 with status `valid`. The 10% Phase 4b scale was 0.8433 ± 0.0868 .

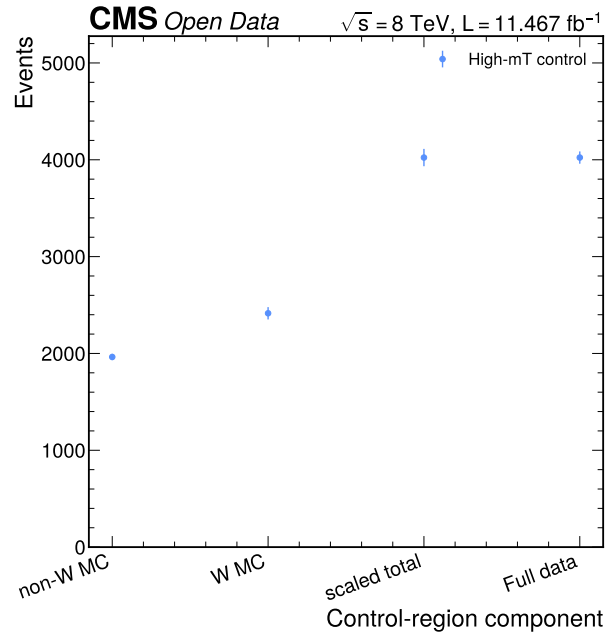


Figure 1: Full high-mT W control comparison. The figure shows non-W MC, nominal W MC, the scaled control-region prediction, and full data in the control region. The scale is derived outside the signal region and then propagated into the observed workspace.

1.3 Full-Data Score-Template Validation

Category	Data	Background	Data/background	Chi2/ndf	Max abs pull
vbf	79	146.67	0.539	4.310	3.34
boosted	2213	2062.80	1.073	1.687	2.14
zero_jet	8466	6894.80	1.228	3.656	2.47

Combined over all score-template bins, data/background is 1.182 with chi2/ndf 3.218. The full-data score-modelling status is **flagged** under the same diagnostic criteria as Phase 4b.

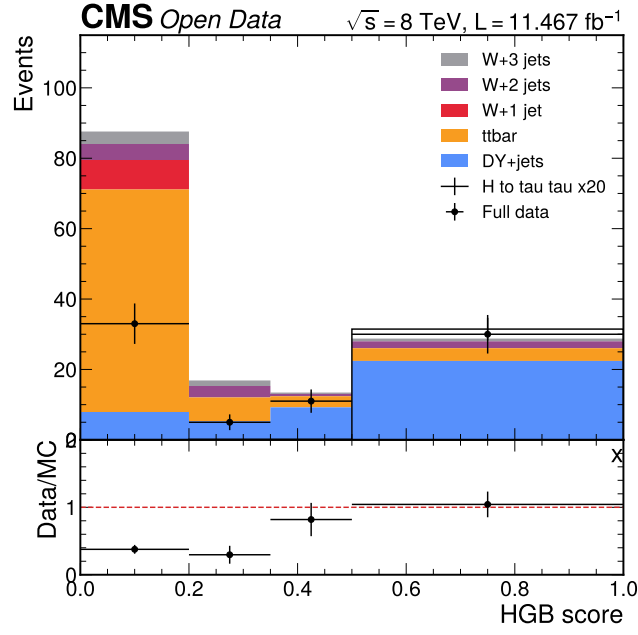


Figure 2: Full-data score-template validation in the VBF category. The plot compares all available Run2012B/C TauPlusX data to the frozen MC template model with the full high- m_T W scale applied. The ratio panel is a diagnostic and does not drive any post-unblinding retuning.

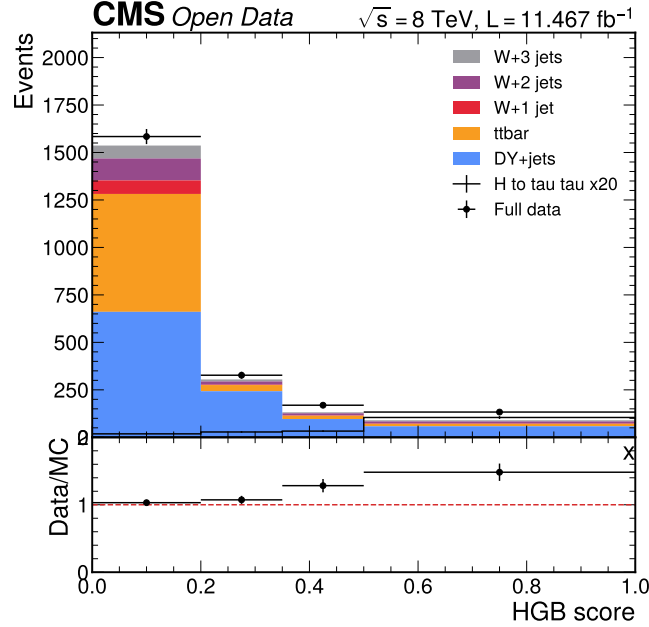


Figure 3: Full-data score-template validation in the boosted category. The plot compares full data to the frozen Phase 4a/4b score-template model. This is an observed-result diagnostic with the Phase 4b score-modeling warning still carried forward.

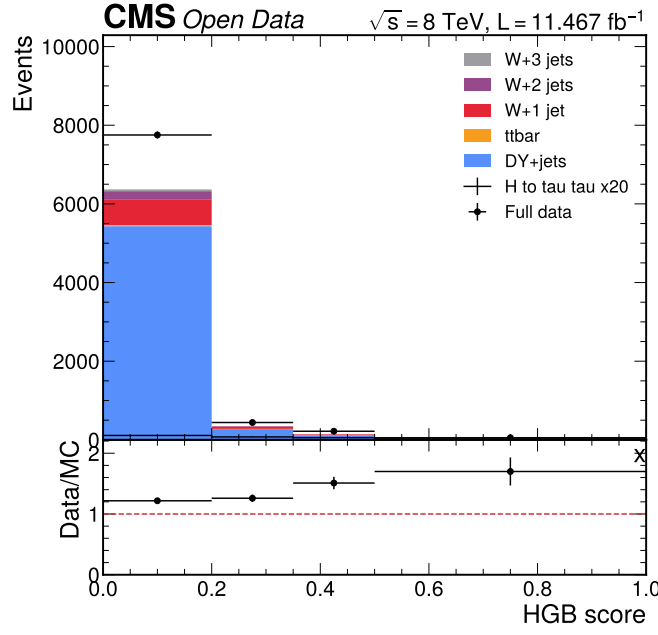


Figure 4: Full-data score-template validation in the zero-jet category. The zero-jet channel dominates the full observed event count and therefore the combined data/MC ratio. The model was not retuned after unblinding.

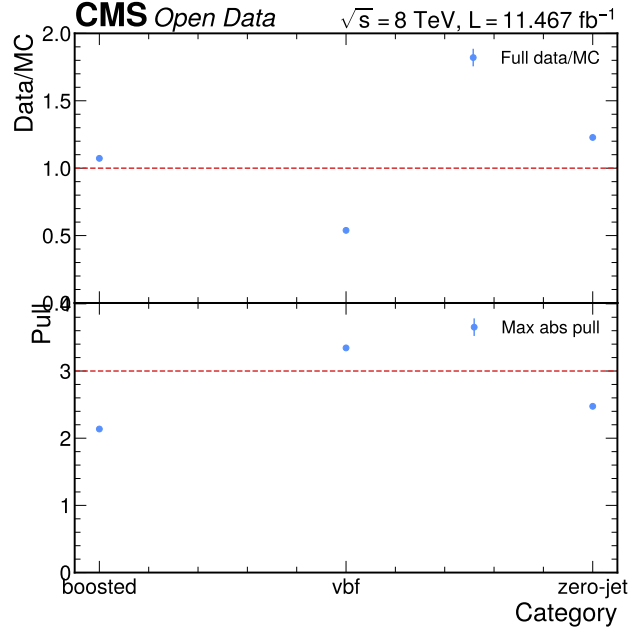


Figure 5: Observed pull and ratio summary. The figure summarizes per-category data/background ratios and maximum score-bin pulls for full data. It compares the observed validation behavior to the Phase 4b 10% validation and Phase 4a expected reference.

1.4 Observed Fit Result

The observed pyhf workspace is written to `pyhf_workspace_observed.json`. Fit status is `evaluated`. The fitted signal strength is 7.6642. The observed 95% CLs upper limit status is `evaluated` and the observed limit is 13.3226 if evaluated. The observed discovery diagnostic status is `evaluated`; its q_0 significance is $Z = 2.9776$ if evaluated.

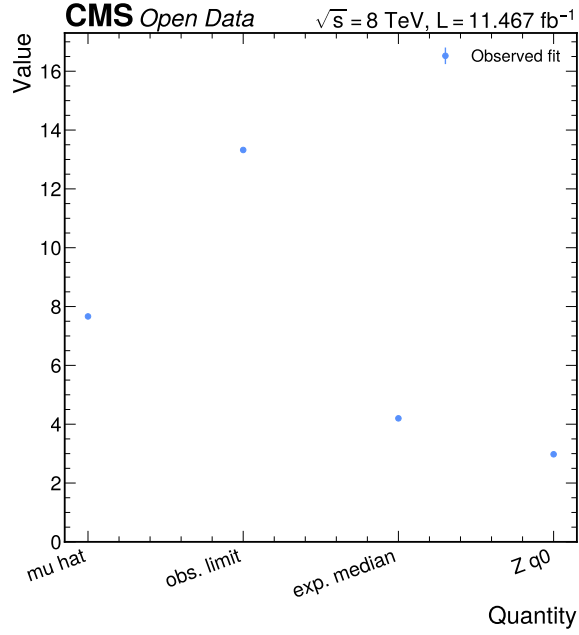


Figure 6: Observed limit and significance summary. The figure shows the full-data fitted signal strength, observed upper limit, Phase 4a expected median limit, and observed discovery diagnostic where available. These numbers are a simplified open-data result and must be interpreted with the documented score-modeling caveat.

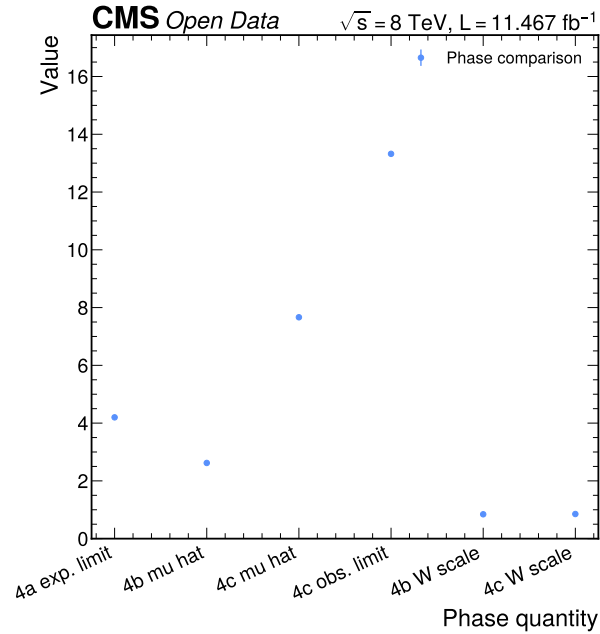


Figure 7: Comparison to Phase 4a expected and Phase 4b validation. The figure compares the Phase 4a expected limit, Phase 4b diagnostic fit, full-data observed fit, and the 10% versus full W high-mT scale factors. It is intended to make the unblinding evolution explicit for Phase 5.

1.5 Phase 5 Obligation

The Phase 5 paper must state that the score-modeling validation was flagged in Phase 4b and remains a limitation of the simplified open-data result. The result should not claim a clean CMS-quality score-template model, nor should it describe the W+jets scale as signal-region tuned.

References